LECTURE 14
MAXIMUM LIKELIHOOD ESTIMATION

Definition

Suppose that the econometrician observes the data $\{W_i : i = 1, \ldots, n\}$, where W_i is a random p-vector. Assume that W_i are iid with the PDF $f(w_i, \theta)$, where $\theta \in \Theta \subset R^k$ is an unknown vector of parameters. The set Θ is usually assumed to be compact.

Example (Normal regression model)**.** Let $W_i = \left(Y_i, X_i^\top\right)^\top$, $\theta = \left(\beta^\top, \sigma^2\right)^\top$, where $\beta \in R^k$ and $\sigma^2 \in R$. Assume that $Y_i = X_i^\top \beta + U_i$, and $U_i \mid X_i \sim N\left(0, \sigma^2\right)$. Then, $f\left(y_i, x_i, \beta, \sigma^2\right) = \left(2\pi\sigma^2\right)^{-1/2} \exp\left(-\left(y_i - x_i^\top \beta\right)^2 / \left(2\sigma^2\right)\right)$.

Since the observations are iid, the joint PDF of $W_1, \ldots, W_n$ is given by

$$\prod_{i=1}^n f(w_i, \theta).$$

The joint PDF gives us the *likelihood* of our sample given the value of θ. The log of the joint PDF is

$$\sum_{i=1}^n \log f(w_i, \theta).$$

We define the log-likelihood function as $1/n$ times the log of the joint PDF evaluated at the random sample $W_1, \ldots, W_n$:

$$\log L_n(\theta) = n^{-1} \sum_{i=1}^n \log f(W_i, \theta).$$

The maximum likelihood (ML) estimator is defined as

$$\widehat{\theta}_n^{ML} = \arg\max_{\theta \in \Theta} \log L_n(\theta).$$

Thus, for a fixed set of observations, the ML estimate is the value of θ for which we are most likely to observe the values of $W_1, \ldots, W_n$ obtained in the sample.

In the normal regression example,

$$\log L_n\left(\beta, \sigma^2\right) = -\frac{1}{2}\log\sigma^2 - \frac{1}{2}\log(2\pi) - \frac{1}{2\sigma^2 n}\sum_{i=1}^n \left(Y_i - X_i^\top \beta\right)^2,$$

and

$$\begin{aligned} \widehat{\beta}_{n,ML} &= \left(\sum_{i=1}^n X_i X_i^\top\right)^{-1} \sum_{i=1}^n X_i Y_i, \\ \widehat{\sigma}^2_{n,ML} &= n^{-1} \sum_{i=1}^n \left(Y_i - X_i^\top \widehat{\beta}_n^{ML}\right)^2. \end{aligned}$$

In this case, the ML estimator of β is identical to the OLS estimator, since maximization of L_n with respect to β is equivalent to minimization of $\sum_{i=1}^n \left(Y_i - X_i^\top \beta\right)^2$.

Asymptotic properties of the ML estimator

Let θ_0 be the true value of θ.

Consistency

By the WLLN, we should expect that for each value of θ,

$$\begin{aligned}
\log L_n(\theta) &= n^{-1}\sum_{i=1}^{n}\log f(W_i,\theta) \\
&\to_p \mathrm{E}[\log f(W_i,\theta)] \\
&= \int (\log f(w,\theta))\, f(w,\theta_0)\, dw.
\end{aligned}$$

Next, consider

$$\begin{aligned}
& \mathrm{E}[\log f(W_i,\theta)] - \mathrm{E}[\log f(W_i,\theta_0)] \\
=\ & \mathrm{E}\left[\log \frac{f(W_i,\theta)}{f(W_i,\theta_0)}\right] \\
\leq\ & \log \mathrm{E}\left[\frac{f(W_i,\theta)}{f(W_i,\theta_0)}\right] \\
=\ & \log \int \frac{f(w,\theta)}{f(w,\theta_0)} f(w,\theta_0)\, dw \\
=\ & \log \int f(w,\theta)\, dw \\
=\ & \log 1 \\
=\ & 0.
\end{aligned}$$

The inequality above follows from the fact that log is a concave function and Jensen's inequality: if f is concave, then $\mathrm{E}[f(X)] \leq f(\mathrm{E}[X])$. The inequality is in fact strict provided that $\Pr(f(W_i,\theta_0) \neq f(W_i,\theta)) > 0$ for all $\theta \neq \theta_0$. As a result, θ_0 uniquely maximizes $\mathrm{E}[\log f(W_i,\theta)]$, and, under additional technical assumptions, we have that

$$\begin{aligned}
\widehat{\theta}_n^{ML} &= \arg\max_{\theta\in\Theta} \log L_n(\theta) \\
&\to_p \arg\max_{\theta\in\Theta} \mathrm{E}[\log f(W_i,\theta)] \\
&= \theta_0.
\end{aligned}$$

Asymptotic normality

The ML estimator solves the first-order conditions

$$0 = n^{-1}\sum_{i=1}^{n}\frac{\partial}{\partial\theta}\log f\left(W_i,\widehat{\theta}_n^{ML}\right).$$

Using the mean value theorem element-by-element, we obtain

$$0 = n^{-1}\sum_{i=1}^{n}\frac{\partial}{\partial\theta}\log f(W_i,\theta_0) + n^{-1}\sum_{i=1}^{n}\frac{\partial^2}{\partial\theta\partial\theta^\top}\log f(W_i,\theta_n^*)\left(\widehat{\theta}_n^{ML}-\theta_0\right), \tag{1}$$

where θ_n^* lies between $\widehat{\theta}_n^{ML}$ and θ_0. Since $\widehat{\theta}_n^{ML} \to_p \theta_0$, we have $\theta_n^* \to_p \theta_0$. Rearranging (1) gives

$$n^{1/2}\left(\widehat{\theta}_n^{ML}-\theta_0\right) = -\left(n^{-1}\sum_{i=1}^{n}\frac{\partial^2}{\partial\theta\partial\theta^\top}\log f(W_i,\theta_n^*)\right)^{-1} n^{-1/2}\sum_{i=1}^{n}\frac{\partial}{\partial\theta}\log f(W_i,\theta_0). \tag{2}$$

Under additional technical assumptions,

$$n^{-1} \sum_{i=1}^{n} \frac{\partial^2}{\partial\theta\partial\theta^\top} \log f(W_i, \theta_n^*) \to_p \mathrm{E}\left[\frac{\partial^2}{\partial\theta\partial\theta^\top} \log f(W_i, \theta_0)\right]. \tag{3}$$

Next, consider $\partial \log f(W_i, \theta_0)/\partial\theta$. Assuming that we can change the order of integration and differentiation,

$$\begin{aligned}
\mathrm{E}\left[\frac{\partial}{\partial\theta} \log f(W_i, \theta_0)\right] &= \mathrm{E}\left[\frac{\partial f(W_i, \theta_0)/\partial\theta}{f(W_i, \theta_0)}\right] \\
&= \int \frac{\partial f(w, \theta_0)/\partial\theta}{f(w, \theta_0)} f(w, \theta_0)\, dw \\
&= \int \frac{\partial f(w, \theta_0)}{\partial\theta} dw \\
&= \frac{\partial}{\partial\theta} \int f(w, \theta_0)\, dw \\
&= \frac{\partial}{\partial\theta} 1 \\
&= 0.
\end{aligned}$$

Thus, by the CLT, we should expect that

$$n^{-1/2} \sum_{i=1}^{n} \frac{\partial}{\partial\theta} \log f(W_i, \theta_0) \to_d N\left(0, \mathrm{E}\left[\frac{\partial}{\partial\theta} \log f(W_i, \theta_0) \frac{\partial}{\partial\theta^\top} \log f(W_i, \theta_0)\right]\right). \tag{4}$$

Combining (2), (3), and (4), we obtain

$$\begin{aligned}
n^{1/2}\left(\widehat{\theta}_n^{ML} - \theta_0\right) &\to_d -\left(\mathrm{E}\left[\frac{\partial^2}{\partial\theta\partial\theta^\top} \log f(W_i, \theta_0)\right]\right)^{-1} N\left(0, \mathrm{E}\left[\frac{\partial}{\partial\theta} \log f(W_i, \theta_0) \frac{\partial}{\partial\theta^\top} \log f(W_i, \theta_0)\right]\right) \\
&= N(0, V),
\end{aligned}$$

where

$$V = \left(\mathrm{E}\left[\frac{\partial^2}{\partial\theta\partial\theta^\top} \log f(W_i, \theta_0)\right]\right)^{-1} \mathrm{E}\left[\frac{\partial}{\partial\theta} \log f(W_i, \theta_0) \frac{\partial}{\partial\theta^\top} \log f(W_i, \theta_0)\right] \left(\mathrm{E}\left[\frac{\partial^2}{\partial\theta\partial\theta^\top} \log f(W_i, \theta_0)\right]\right)^{-1}.$$

Next,

$$\begin{aligned}
\mathrm{E}\left[\frac{\partial^2}{\partial\theta\partial\theta^\top} \log f(W_i, \theta_0)\right] &= \mathrm{E}\left[\frac{\partial}{\partial\theta^\top} \frac{\partial f(W_i, \theta_0)/\partial\theta}{f(W_i, \theta_0)}\right] \\
&= \mathrm{E}\left[\frac{\partial^2 f(W_i, \theta_0)/\partial\theta\partial\theta^\top}{f(W_i, \theta_0)}\right] - \mathrm{E}\left[\frac{\partial f(W_i, \theta_0)/\partial\theta}{f(W_i, \theta_0)} \frac{\partial f(W_i, \theta_0)/\partial\theta^\top}{f(W_i, \theta_0)}\right] \\
&= -\mathrm{E}\left[\frac{\partial}{\partial\theta} \log f(W_i, \theta_0) \frac{\partial}{\partial\theta^\top} \log f(W_i, \theta_0)\right],
\end{aligned} \tag{5}$$

since

$$\begin{aligned}
\mathrm{E}\left[\frac{\partial^2 f(W_i, \theta_0)/\partial\theta\partial\theta^\top}{f(W_i, \theta_0)}\right] &= \int \frac{\partial^2 f(w, \theta_0)/\partial\theta\partial\theta^\top}{f(w, \theta_0)} f(w, \theta_0)\, dw \\
&= \int \frac{\partial^2 f(w, \theta_0)}{\partial\theta\partial\theta^\top} dw \\
&= \frac{\partial^2}{\partial\theta\partial\theta^\top} \int f(w, \theta_0)\, dw \\
&= \frac{\partial^2}{\partial\theta\partial\theta^\top} 1 \\
&= 0.
\end{aligned}$$

The result in (5) is called the *information equality*. It implies that

$$\begin{aligned} V &= -\left(\mathrm{E}\left[\frac{\partial^2}{\partial\theta\partial\theta^\top}\log f\left(W_i,\theta_0\right)\right]\right)^{-1} \\ &= I^{-1}\left(\theta_0\right), \end{aligned}$$

where

$$I\left(\theta_0\right) = -\,\mathrm{E}\left[\frac{\partial^2}{\partial\theta\partial\theta^\top}\log f\left(W_i,\theta_0\right)\right]$$

is called the *information matrix.* Thus, we have that

$$n^{1/2}\left(\widehat{\theta}_n^{ML}-\theta_0\right)\to_d N\left(0, I^{-1}\left(\theta_0\right)\right).$$

Remarks

- In general, ML estimation requires numerical maximization of the log-likelihood function.
- Hypothesis testing can be performed using the Wald-type statistic. Suppose that $H_0 : h\left(\theta_0\right) = 0$, where $h : R^k \to R^q$. The Wald statistic is given by

$$W_n = nh\left(\widehat{\theta}_n^{ML}\right)^\top \left(\frac{\partial h\left(\widehat{\theta}_n^{ML}\right)}{\partial\theta^\top} I^{-1}\left(\widehat{\theta}_n^{ML}\right)\frac{\partial h^\top\left(\widehat{\theta}_n^{ML}\right)}{\partial\theta}\right)^{-1} h\left(\widehat{\theta}_n^{ML}\right).$$

  One should reject the null if $W_n > \chi^2_{q,1-\alpha}$. Alternatively, the null hypothesis $h\left(\theta_0\right) = 0$ can be tested using the *likelihood ratio* statistic

$$LR_n = 2\left(\log L_n\left(\widehat{\theta}_n^{ML}\right) - \log L_n\left(\widetilde{\theta}_n^{ML}\right)\right),$$

  where $\widetilde{\theta}_n^{ML}$ is the null-restricted ML estimator:

$$\widetilde{\theta}_n^{ML} = \arg\max_{\theta\in\Theta, h(\theta)=0} \log L_n(\theta).$$

  Under the null, $LR_n \to_d \chi^2_q$, and is asymptotically equivalent to the Wald statistic.
- The ML estimator is efficient. Let $\widehat{\theta}_n$ be an estimator such that $n^{1/2}\left(\widehat{\theta}_n-\theta_0\right)\to_d N\left(0,\Sigma\left(\theta_0\right)\right)$. Then $\Sigma\left(\theta_0\right) - I^{-1}\left(\theta_0\right)$ is positive semi-definite. Thus, the asymptotic variance of the ML estimator is as small as, or smaller than, the variance of any consistent and asymptotically normal estimator.
- The ML estimation relies on very strong assumptions: the true PDF is known up to the value of parameters. If the PDF is misspecified, the estimator is called the quasi-ML estimator. In some cases, the quasi-ML estimator is still consistent. The OLS estimator is an example. It is still consistent even if the data are not normally distributed.